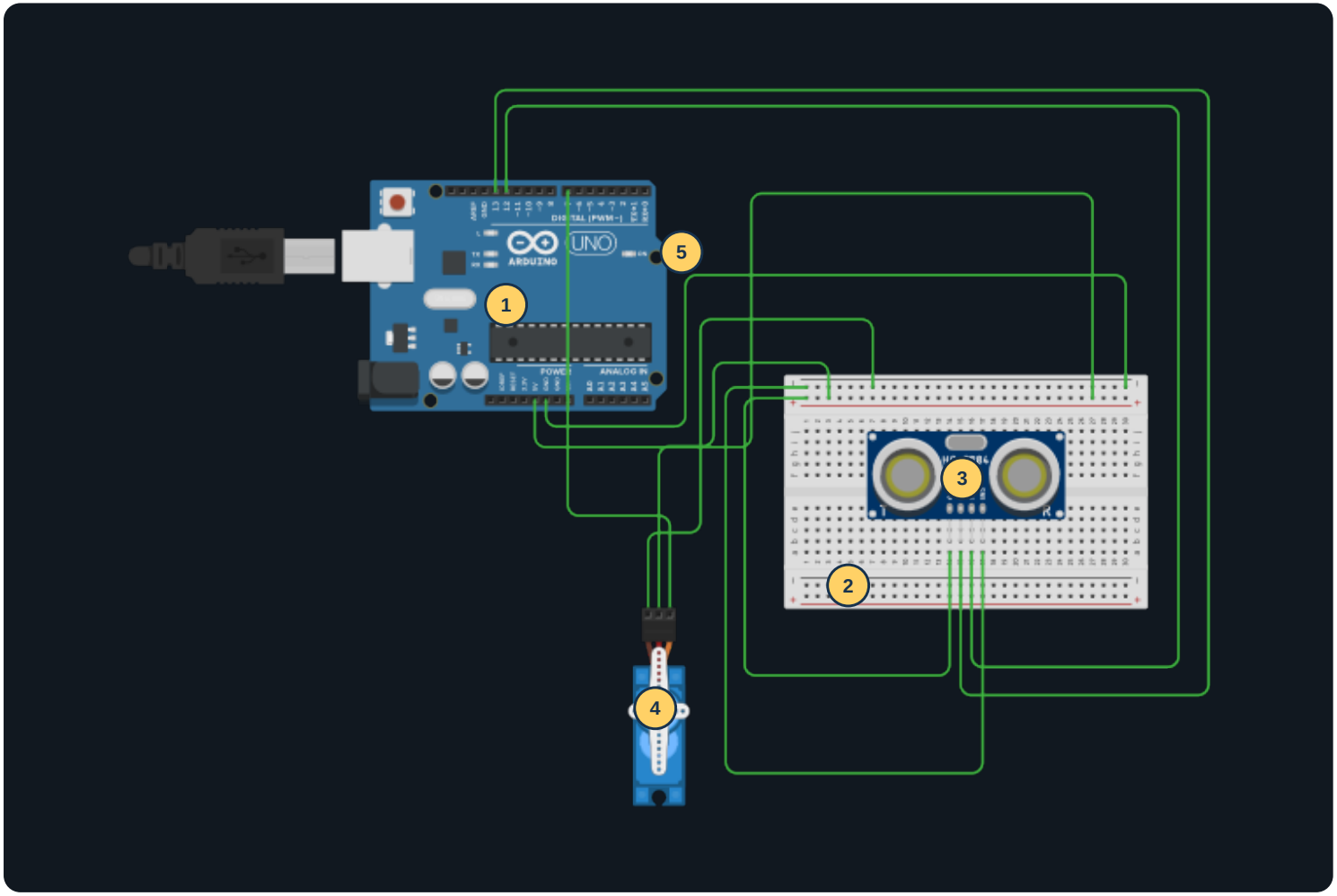
01 | Ultrasonic Distance Sensor + Servo

Arduino Uno reads distance from an HC-SR04 and drives a small servo.



COMPONENT KEY

1. **Arduino Uno R3**
   Microcontroller board; runs the circuit program.
2. **Breadboard**
   Solderless board for temporary connections.
3. **HC-SR04 ultrasonic sensor**
   Measures distance using a trigger pulse and echo return.
4. **Micro servo**
   Position-controlled actuator; exact model is not shown.
5. **Jumper wires**
   Carry power and signals between components.

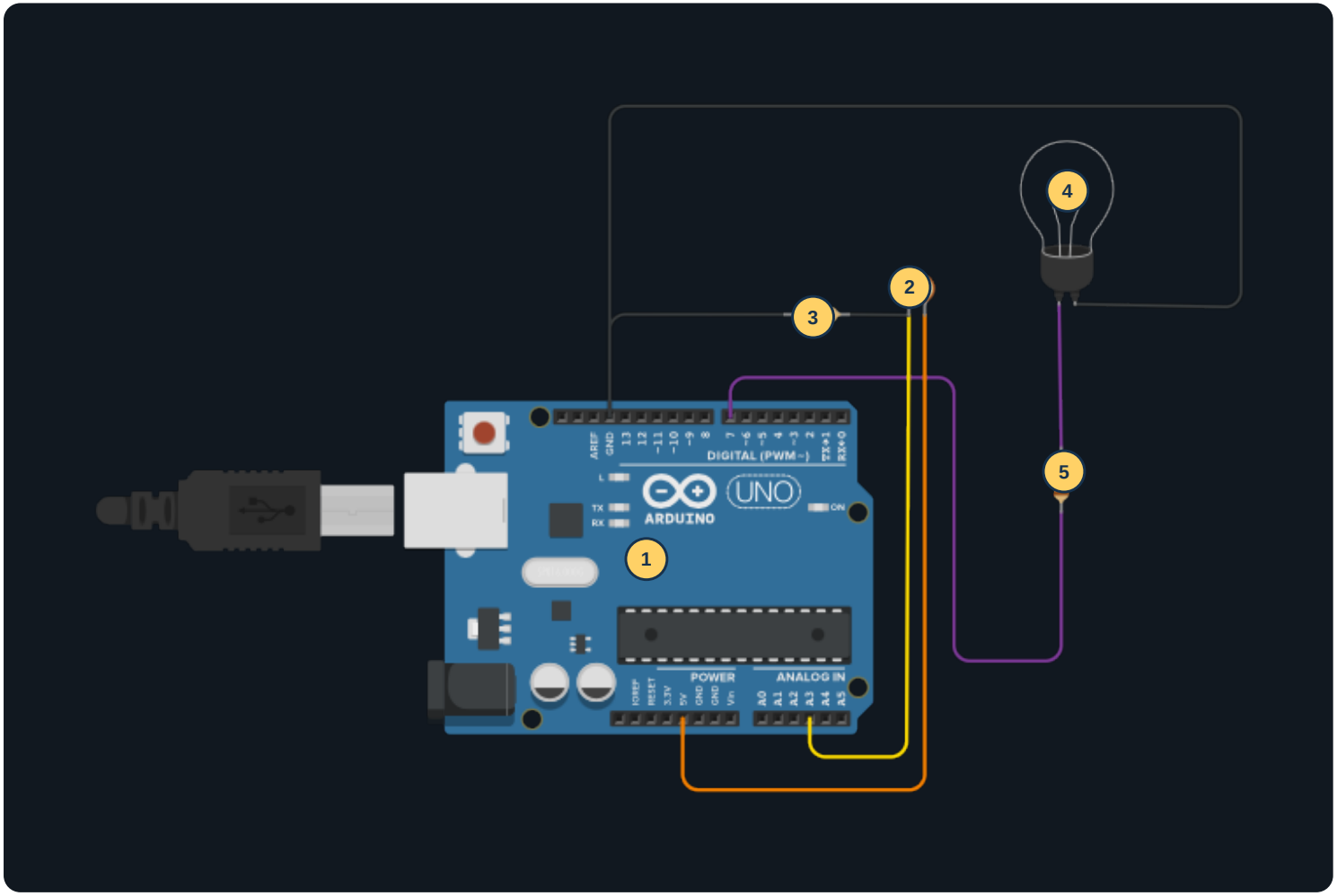
Component numbering matches the badges on the diagram. Exact resistor values and model numbers are not readable in the supplied images.

Wiring colors follow the source illustration; confirm pin assignments against the project code.

Identification is based on visible shapes and markings. The diagram images do not show the source code or a complete pin-by-pin wiring list.

02 | Light Sensor + Lamp

An LDR changes resistance with light; the Arduino reads the sensor and controls a lamp.



COMPONENT KEY

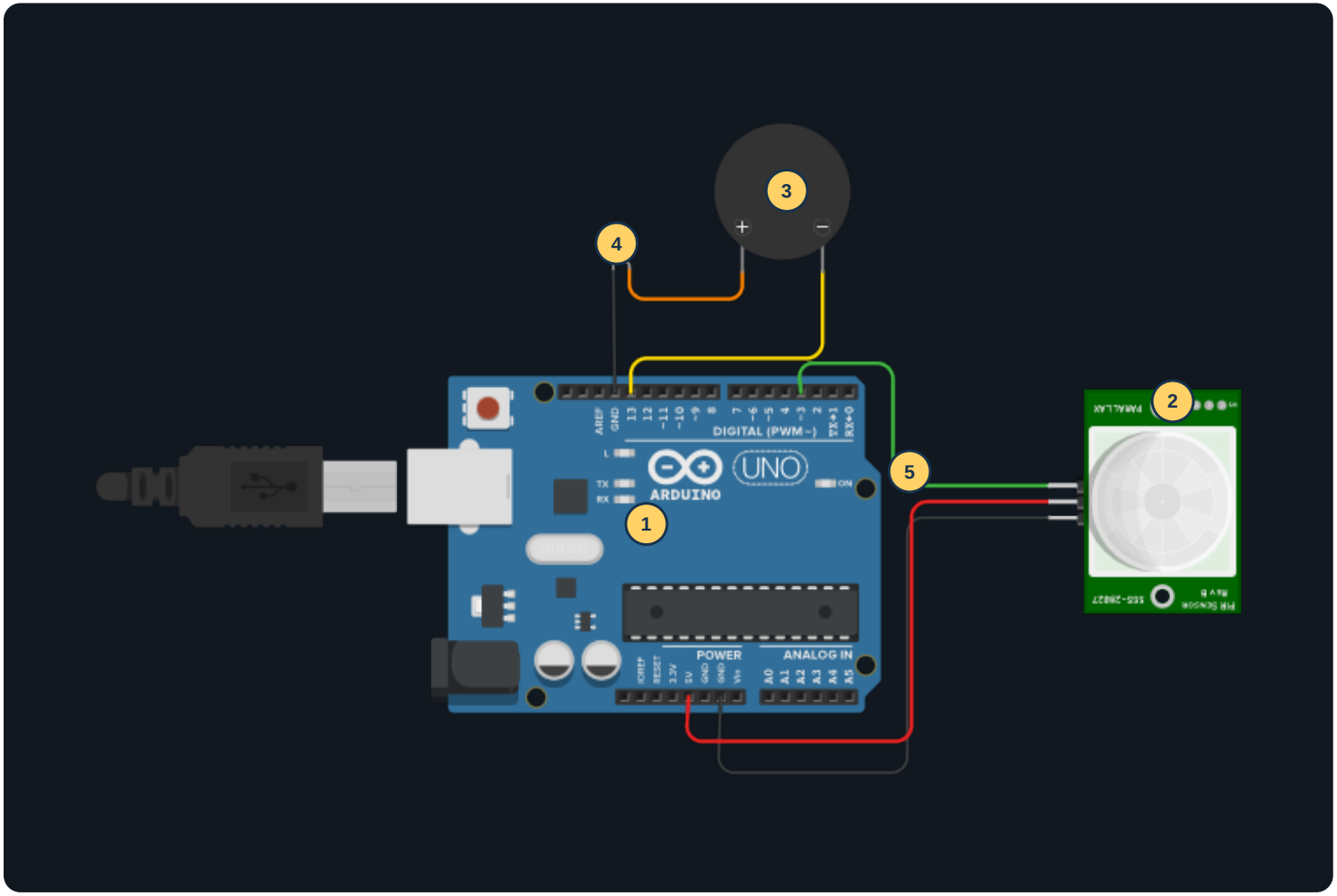
1. **Arduino Uno R3**
   Microcontroller board; reads the light-sensor voltage.
2. **Photoresistor (LDR)**
   Light-dependent resistor used to sense brightness.
3. **Fixed resistors (2)**
   Limit or divide current/voltage; values are not legible.
4. **Lamp / light bulb**
   Output load shown in the diagram.
5. **Jumper wires**
   Connect the sensor, lamp, and Arduino.

Component numbering matches the badges on the diagram. Exact resistor values and model numbers are not readable in the supplied images.

Wiring colors follow the source illustration; confirm pin assignments against the project code.

03 | PIR Motion Sensor + Buzzer

A PIR module detects motion; the Arduino signals with a buzzer and indicator LED.



COMPONENT KEY

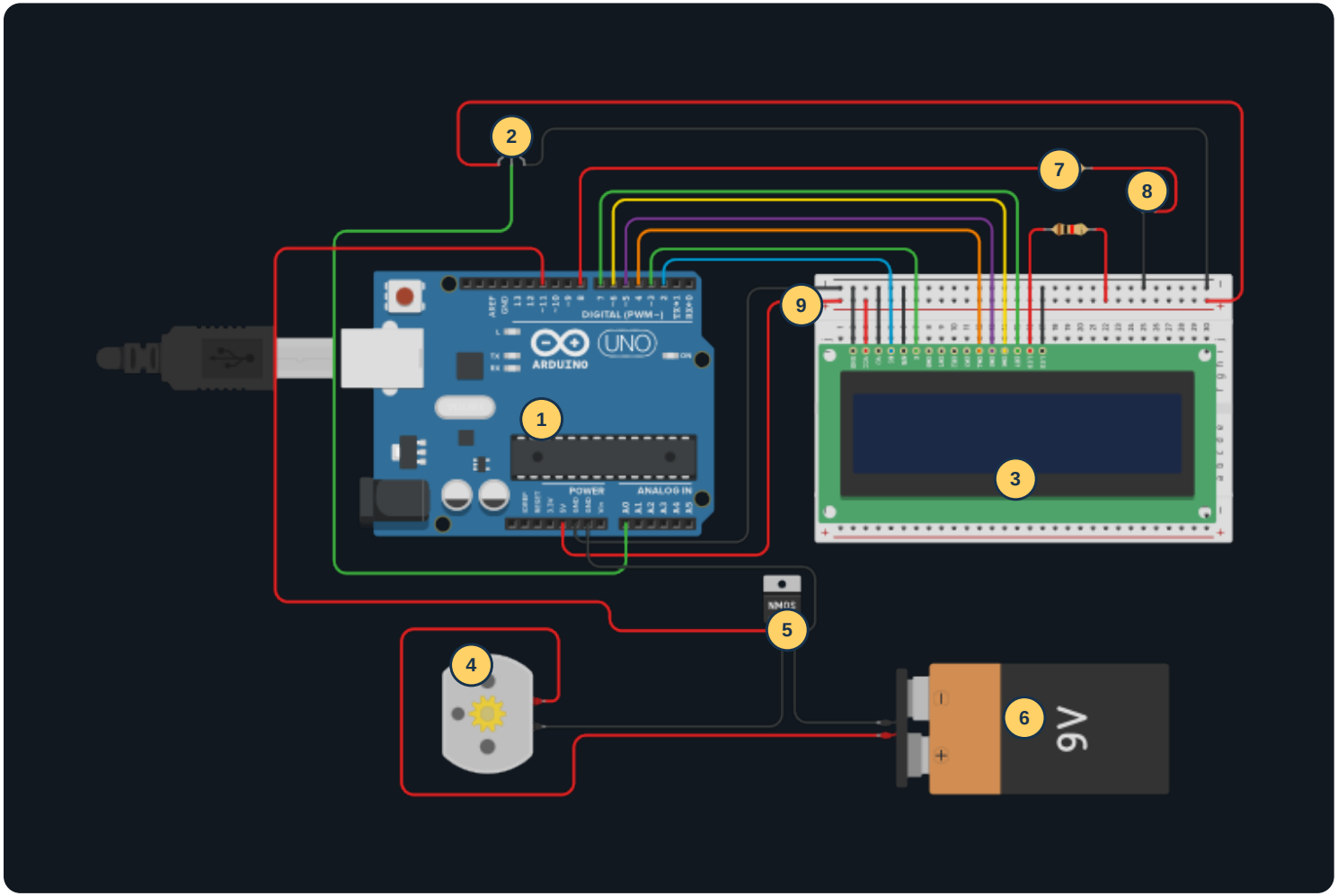
1. **Arduino Uno R3**
   Microcontroller board; processes the motion signal.
2. **PIR motion sensor module**
   Passive infrared sensor; module markings suggest an HC-SR501 style.
3. **Piezo buzzer**
   Audible alert output.
4. **LED indicator**
   Visual status or alert output.
5. **Jumper wires**
   Carry power and signal connections.

Component numbering matches the badges on the diagram. Exact resistor values and model numbers are not readable in the supplied images.

Wiring colors follow the source illustration; confirm pin assignments against the project code.

04 | Temperature-Controlled Fan + LCD

A temperature sensor feeds the Arduino, which displays information and switches a fan motor.



COMPONENT KEY

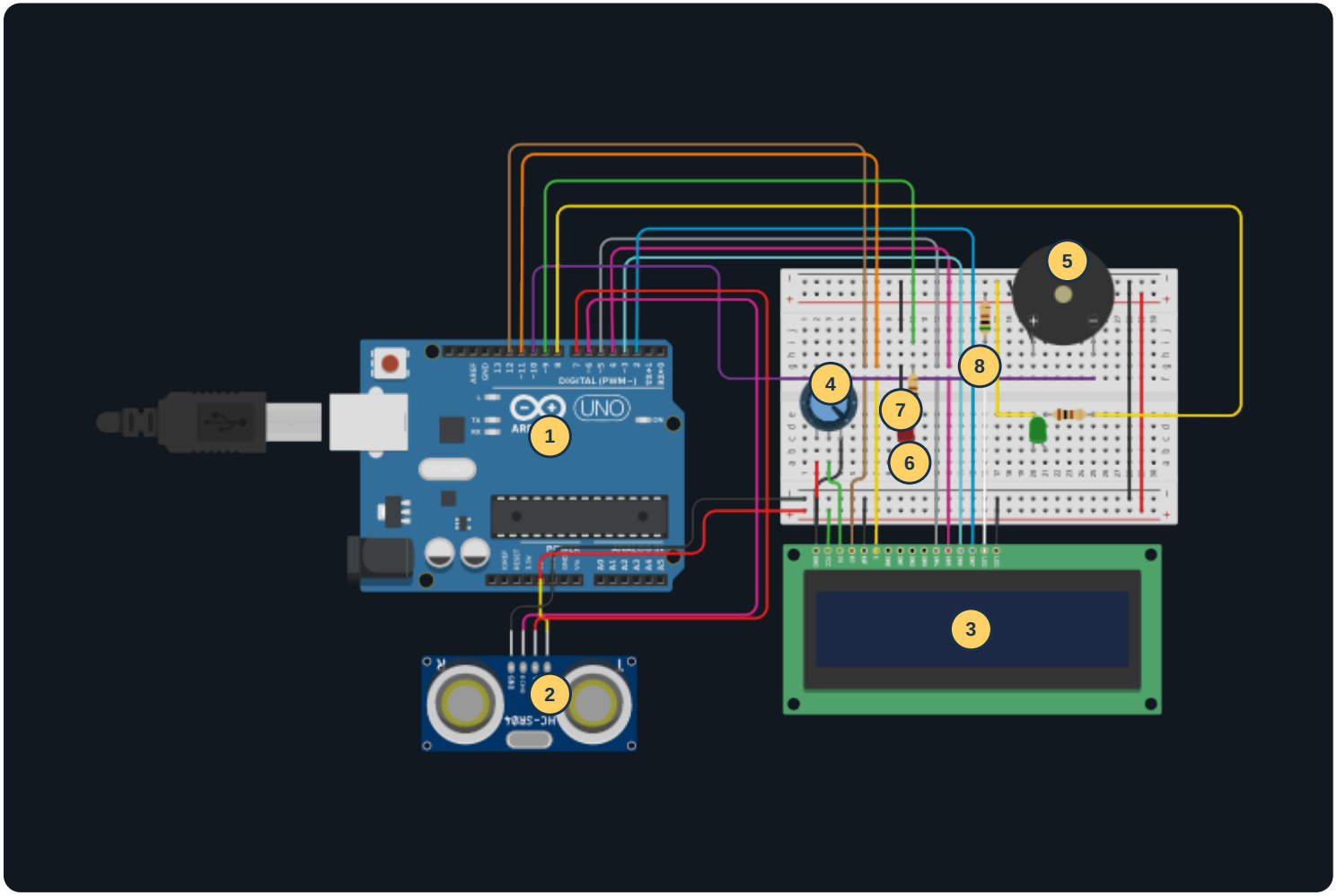
1. **Arduino Uno R3**
   Microcontroller board; reads temperature and controls outputs.
2. **TMP temperature sensor**
   Three-pin analog sensor marked “TMP”; exact part number is unclear.
3. **16×2 character LCD**
   Displays two rows of text.
4. **DC motor / fan**
   Cooling output load.
5. **N-channel MOSFET**
   Transistor switch for the motor load.
6. **9 V battery**
   Separate motor supply shown; confirm polarity and ratings before wiring.
7. **Fixed resistors (2)**
   Circuit resistors; values are not legible.
8. **LED indicator**
   Visual status output.
9. **Breadboard + jumper wires**
   Temporary assembly and interconnections.

Component numbering matches the badges on the diagram. Exact resistor values and model numbers are not readable in the supplied images.

Wiring colors follow the source illustration; confirm pin assignments against the project code.

05 | Ultrasonic Range Display + Alert

An HC-SR04 distance reading is shown on an LCD, with LEDs and a buzzer for status alerts.



COMPONENT KEY

1. **Arduino Uno R3**
   Microcontroller board; measures and reports distance.
2. **HC-SR04 ultrasonic sensor**
   Measures distance from trigger/echo timing.
3. **16×2 character LCD**
   Displays the measured distance or status.
4. **Potentiometer**
   Adjustable resistor, likely for LCD contrast.
5. **Piezo buzzer**
   Audible alert output.
6. **Red + green LEDs**
   Visual indicators; exact status meanings depend on the code.
7. **Fixed resistors**
   Current-limiting / signal resistors; values are not legible.
8. **Breadboard + jumper wires**
   Temporary assembly and interconnections.

Component numbering matches the badges on the diagram. Exact resistor values and model numbers are not readable in the supplied images.

Wiring colors follow the source illustration; confirm pin assignments against the project code.